

SUBMITTED BY: AYESHA IQBAL

ASSIGNMENT # 1

HR ANALYTICS

SUBMITTED TO: ROHIT NISHANT

Contents

1 CONTEXT....1
1.1 The Goal of the Analysis....1
1.2 Data Quality Issues....1
1.3 WORKFORCE DEMOGRAPHICS....1
2 Conflict....2
2.1 Attrition in relation to Department....2
2.2 Attrition in relation to Gender and Age....3
2.3 Attrition by Educational Level....4
2.4 Attrition and Salary....4
2.5 Attrition in Relation to Job Satisfaction and Work life Balance....5
2.6 Attrition in Relation to Overall Satisfaction in a Job Role....6
3 RESOLUTION:....6
3.1 Retention Rate by Job Satisfaction....6
3.2 Work-Life Balance Impact on Attrition....7
Recommendations....8
3.3 Attrition rate by Department:....8
3.4 Average Salary of Employees Who Left and Stayed....9
4 Conclusion....9
5 REFERENCE....9

1 CONTEXT

The Company is experiencing a 19% employee attrition rate, raising concerns regarding employee retention. An attrition rate quantifies turnover within a specified timeframe. The attrition rate directly affects productivity, operational efficiency, and profitability. Recognizing the determinants of employee turnover is essential for cultivating a stable and motivated team. The company intends to leverage its current data to investigate the primary factors contributing to this phenomenon. It had previously used Excel for reporting and wants to use Tableau for more extensive, data-driven decision-making and analysis.

To comprehend the factors contributing to the rise in the attrition rate, we will conduct an analysis utilizing Tableau's dashboarding tools. Various factors can contribute to an elevated attrition rate, such as job satisfaction, workplace environment, salary hikes, work-life balance, etc., frequently regarded as hygiene factors by various human resource specialists.

1.1 The Goal of the Analysis

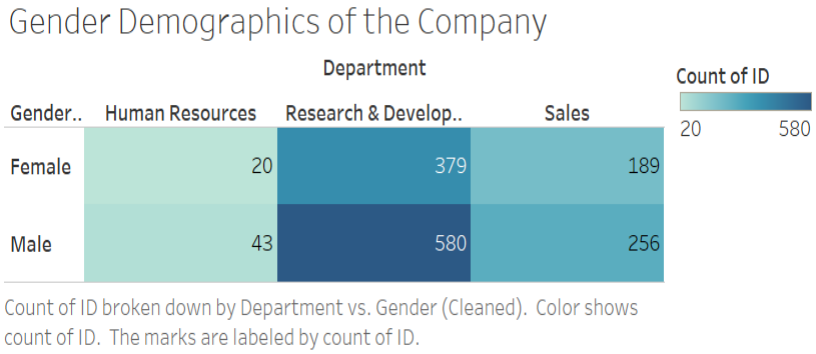
The objective is to ascertain all factors influencing the rise in the attrition rate through visualization and dashboarding methodologies. To establish a correlation between these variables and attrition, and subsequently derive solutions to the issue while formulating recommendations through this analysis.

1.2 Data Quality Issues

All the data quality issues were found in Age, Gender, and Date of Birth, which were resolved in Excel and some in Tableau, using calculated fields to standardise inconsistent values.

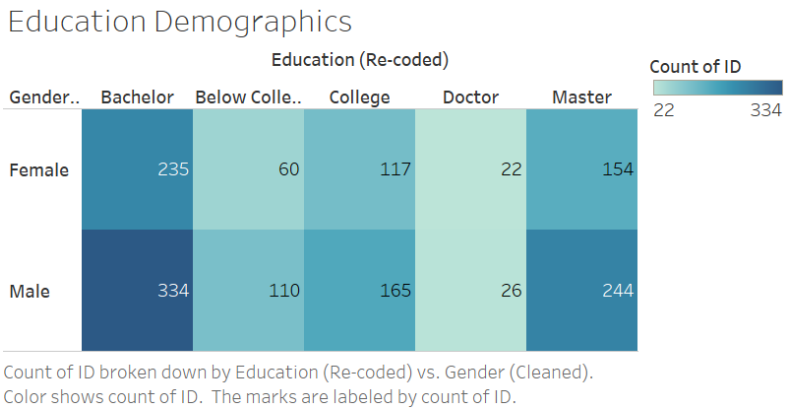
1.3 Workforce Demographics

The company has an overall workforce of 1467 employees. The company employs 588 females and 879 males. The figure displays the gender distribution department view as follows:



The data provided relates to three primary departments of the company. The Research and Development Department, Human Resources Department, and Sales Department. The age of the staff members ranges from 18 to 85 years. The average age for the Human Resource department is 38.68, for Research and Development it is 37.064, and for the Sales Department, it is 36.515.

The educational demographics differ among various departments and between the two genders. We have categorized education into five segments: below college, college, bachelor's, master's, and doctoral/PhD. The total number of employees is as follows: 170 with below-college education, 282 at the college level, 569 with bachelor's degrees, 398 with master's degrees, and 49 with doctoral degrees. These categories have been further subdivided into distribution tables based on different factors. Table 2 presents the count of education levels categorized by gender, while Table 3 illustrates the count of education levels by department.



2 CONFLICT

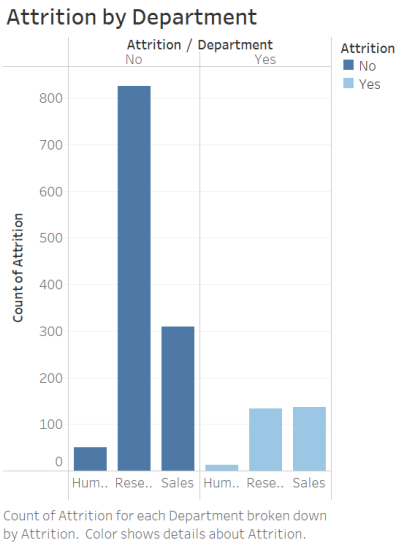
The initial analysis suggests an attrition rate of 19.15%. An increase in attrition is costing the company intellectual capital, expertise, recruitment and training costs, employer brand damage, and most importantly, employee demotivation, etc., which is alarming for the upper management in the company. By analyzing the number of employees who left compared to those who remained, we may investigate the basic factors influencing attrition.

We will visualize several pertinent variables in relation to our target variable to identify which significantly impacts the attrition rate.

2.1 Attrition in relation to Department

First, we will check the attrition rate across different demographics, starting with the department's attrition rate.

The figure indicates that the Sales Department has the highest attrition count, with 136, followed closely by the R&D Department, which has 133. The minimum is observed in the HR Department, at 12.



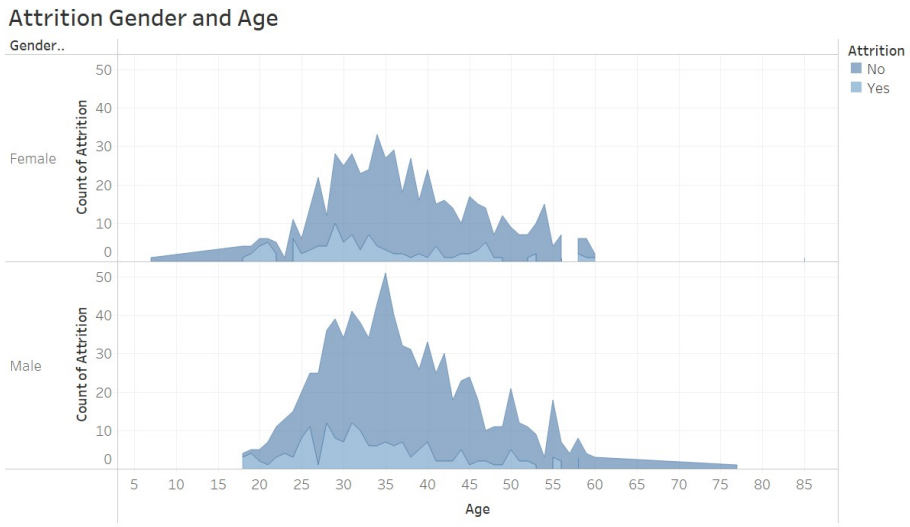
Key Takeaways

- The high attrition rate in the sales and R&D departments may indicate a lack of career development and growth opportunities, a performance-driven environment, and unsatisfactory job roles.
- Low attrition in HR may indicate stable roles, effective policies, or higher satisfaction.

2.2 Attrition in relation to Gender and Age

The trend shows that employees in their late 20s and early 30s leave the company at increased rates. This trend is evident in both genders, although it is more pronounced in males. Younger employees may encounter challenges, including role dissatisfaction, restricted career development options, or issues related to work-life balance, resulting in attrition. In contrast, we see that the attrition rate diminishes with age. Senior employees exhibit a lower tendency to leave the company, indicating enhanced employment security and satisfaction within this demographic. Another observation in this analysis is that, although attrition patterns for males and females are similar, males exhibit a more pronounced peak at younger ages, suggesting that efforts to retain younger male employees may require special attention.

An anomaly is observed in females around 60, potentially attributable to retirement preparations or health-related concerns within this demographic.



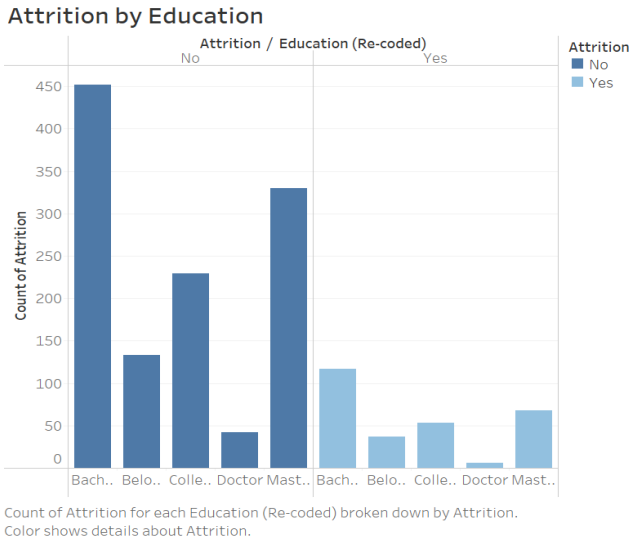
The plot of count of Attrition for Age broken down by Gender (Cleaned). Color shows details about Attrition.

Key Takeaways

- To retain younger employees, there needs to be work on promoting career development, mentorship, and clear advancement paths.
- Investigation is required to determine the causes of younger employees' attrition rate.

2.3 Attribution by Educational Level

The Bar chart indicates that employees with Bachelor's degrees exhibit the highest attrition rate, followed by those with Master's degrees, while Doctoral degree holders demonstrate the lowest attrition rate. This may coincide with our previous finding, i.e., the age bracket of 20–30 demonstrates a higher attrition rate, while older ones show stability. In contrast to our findings, Dr. L. Kesavan (2022) through her research proved “that education and other demographics had little to no effect on attrition, with f statistics 0.984 and p-value greater than 5% in an Anova test, it proves to have insignificant relation.”

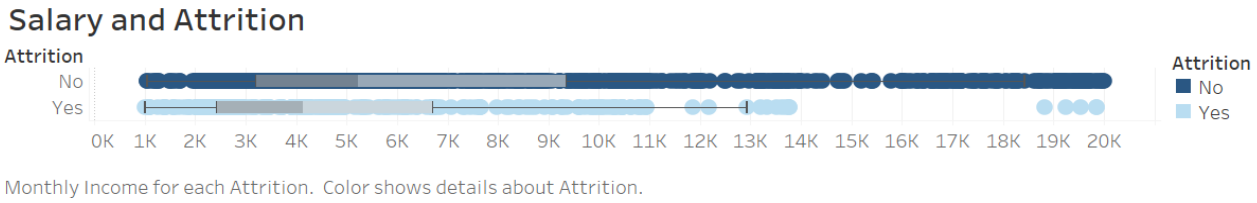


Key Takeaways:

- The organizational culture may not align well with the younger mindset. They might be inclined towards a more innovative, inclusive, or flexible work environment.
- Many employees in this category have student loans to pay off. They might be attracted to higher-paying jobs for this purpose. The company needs to match the competitor's pace in this regard.

2.4 Attrition and Salary

The visualization depicts a view of salary ranges for those who stayed and those who are showing an intention to leave. Those who showed an intention to leave had a lower median monthly income in comparison to those who stayed. The salary range for the first group (no) indicates



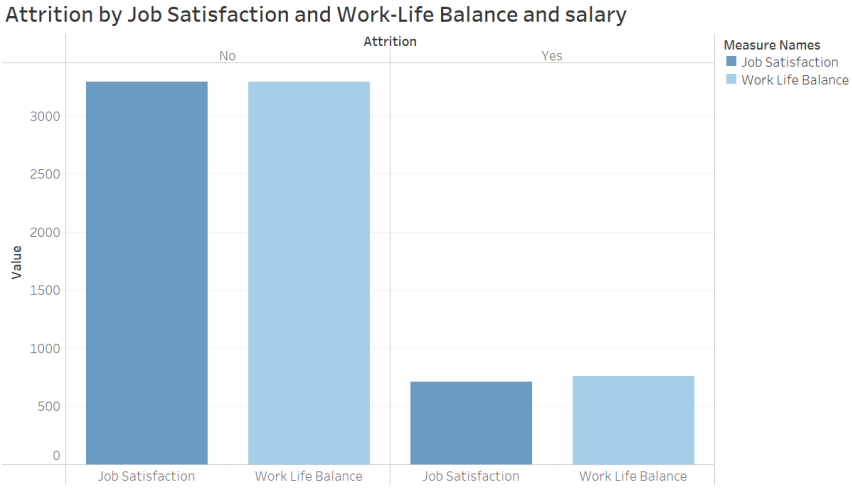
greater variety. This may result from causes such as promotions, salary increases, or varying job levels. The data also showed a few outliers, but there might be a few exceptional cases. Hence, we can assume that salary has a strong relation with attrition rate.

Key Takeaways:

- Salary structures and compensation policies need reformation.
- There are visible pay disparities within the organization.

2.5 Attrition in Relation to Job Satisfaction and Work life Balance

These metrics are, although quite different, directly proportional to one another. Rajasekar and Venkateswara Prasad (2017) found that 82% of respondents would quit if work interfered with their personal life. The balance between work and personal life leads to job satisfaction, indicating a positive correlation."



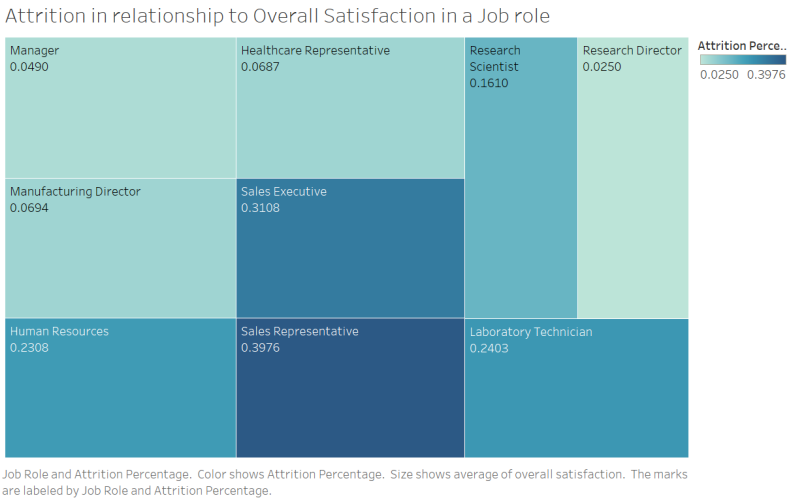
Our visualization indicates that employees who stayed with the company exhibited notedly higher job satisfaction and work-life balance as compared to those who left. Those who stayed showed content in these categories expressing improved happiness and balance during their tenure.
However, those who left had lower job satisfaction and work-life balance. These metrics drop significantly, suggesting they may serve as predictors for attrition.
In both attrition groups, work-life balance is marginally better than job satisfaction. It appears that employees may prioritize work-life balance over job satisfaction when leaving.

Key Takeaways:

- Both factors significantly influence attrition and retention. Employees with better work-life balance and job satisfaction quit less often.
- Our data suggests employees value work-life balance over job satisfaction.

2.6 Attrition in Relation to Overall Satisfaction in a Job Role

The tree map indicates that Sales Representatives and Sales Executives have a high turnover rate, indicating low job satisfaction. Healthcare Representatives and Research Scientists have lower attrition rates. These roles have higher employee satisfaction and lower attrition.



In contrast to the cases mentioned, Human Resources and laboratory technicians offer a divergent perspective. Having a moderate overall satisfaction rate, they exhibit a

moderate attrition rate as well, indicating a mixed reaction from employees. This situation can be resolved through timely intervention.

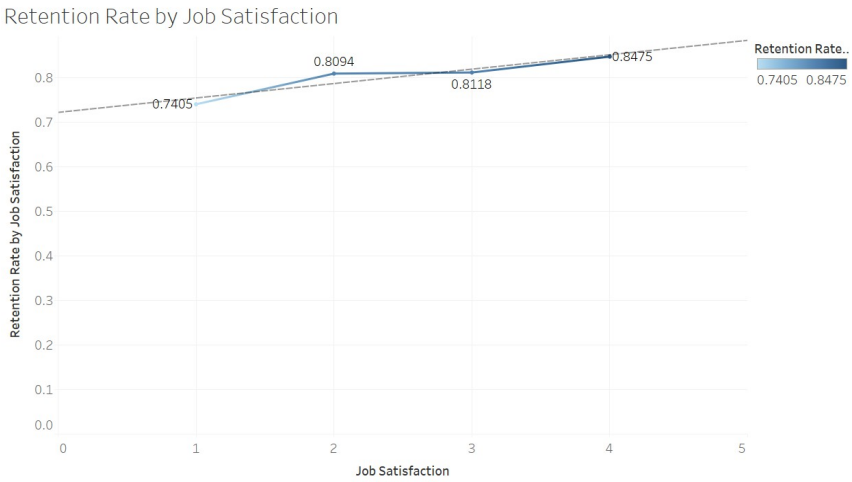
Key Takeaways:

- Dissatisfaction is notably visible in Sales roles, which indicates job dissatisfaction that may include a toxic work environment, excessive work pressure, lack of work-life balance, incentive-related concerns, etc.
- There's a strong correlation between satisfaction and retention.

3 RESOLUTION

In this section, we will discuss the solutions that the company can implement to reduce the attrition rate.

3.1 Retention Rate by Job Satisfaction



The trend of Retention Rate by Job Satisfaction for Job Satisfaction. Color shows Retention Rate by Job Satisfaction. The marks are labeled by Retention Rate by Job Satisfaction.

The graph shows a steady increase in retention rate with an increase in job satisfaction. Here we have divided job satisfaction into multiple levels, where 1 represents “very low," 2 for “medium," 3 for “high,” and 4 for “very high." At 1, the retention rate is around 74.05%, which with each increasing level rises to the highest level 4 and is 84.75%. The largest increase in retention rate occurs between job satisfaction levels 1 and 2, suggesting that sometimes even the smallest of improvements in job satisfaction may lead to a drastic improvement in retention.

R-Squared =0.8714

Job satisfaction accounts for approximately 87.14% of the variation in retention rate. However, this model is fitted on only four satisfaction levels, so the figure should be read as a clear upward trend rather than strong evidence of explanatory power.

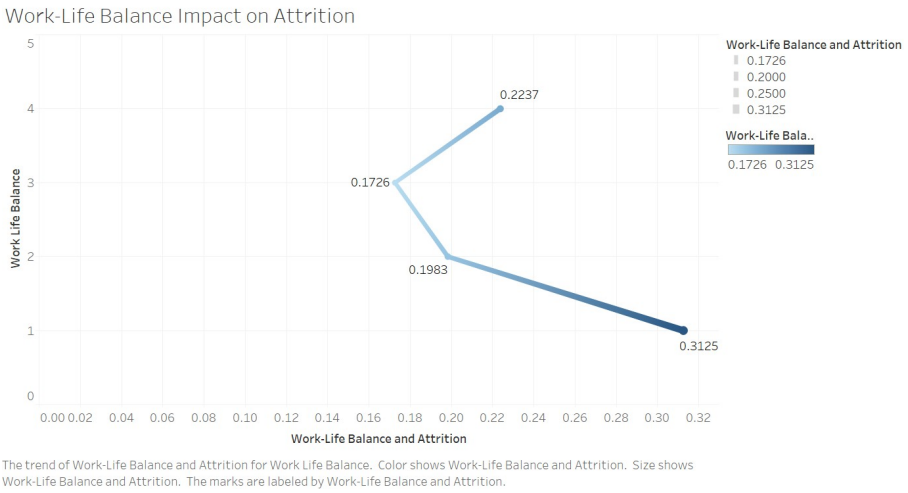
P-Value (Significance): p=0.0665

The model's p-value is just above 0.05, so the relationship is not statistically significant at the 5% level. A model built on individual employee records, rather than four aggregated levels, would give a more reliable test.

Recommendations

- The company should target dissatisfied employees. Dealing with low satisfaction early may reduce attrition. Identify and address dissatisfied employees' needs (1 and 2). One-on-one interviews or surveys may reveal dissatisfaction drivers.
- For the employees showing higher levels of satisfaction, i.e., (3 and 4), we need to formulate strategies focused on retaining their engagement. These actions could include skill update training, growth opportunities, rewards and incentives, a recognition program, and mainly a positive work environment.
- Tools like pulse surveys can be used to monitor satisfaction trends and proactively intervene when needed.

3.2 Work-Life Balance Impact on Attrition



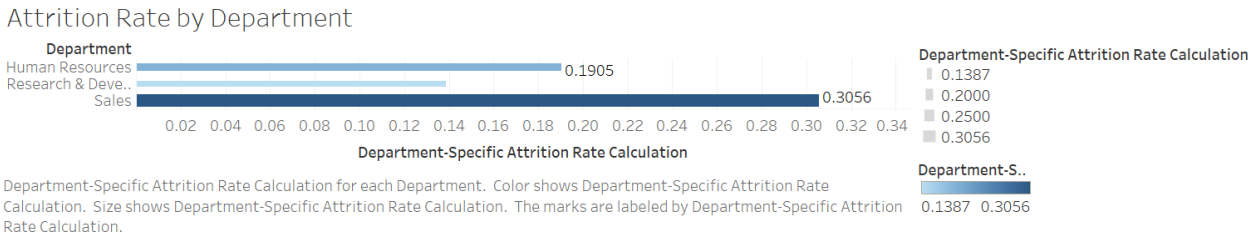
According to Sankararaman Ganapathyraman et al. (2023), work-life balance is the main cause of attrition, with a 27% correlation. To determine whether it's accurate, we analyze its impact on attrition. This graph represents the relationship between work-life balance and attrition. Attrition and work-life balance are on the x and y axes, respectively. Higher y-axis scores indicate better work-life balance. The work-life balance is divided into multiple levels. Level 1 means low; at this point, the attrition rate is highest, i.e., 31.25%. Level 2 scales medium at work-life balance, and at this point, there's a sharp decline in attrition, i.e., 19.83%. Improvement in work-life balance from 1 to 2 strongly affected attrition. At Level 3, a slight elevation in attrition to 22.37% is observed, which suggests that further improvements may not influence attrition as strongly as observed previously. At Level 4, attrition drops to 17.26%, the lowest. This indicates employees with high levels of work-life balance show less propensity towards leaving the company.

Recommendations

- Staff with poor work-life balance should be the company’s top priority. Interventions focused on Level 1 could significantly improve the retention rate.
- Maintaining positive relationships with Level 4 employees will reduce attrition. Create employee-friendly and work-life-balanced strategies to build trust.

3.3 Attrition rate by Department:

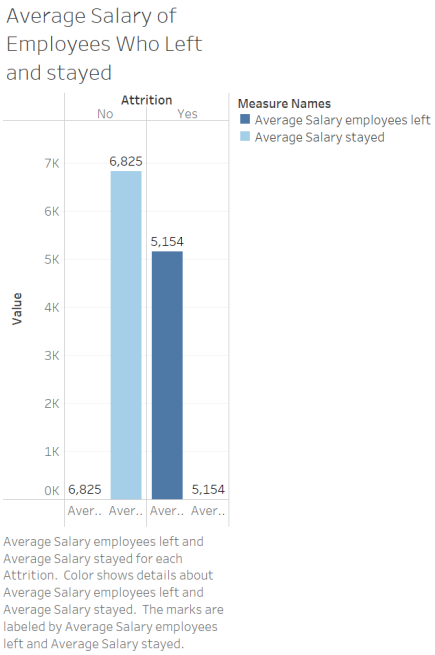
This chart depicts the attrition rate by department; this can serve as a tool for predicting which departments need more attention in terms of retention efforts. The R&D department scores the lowest on attrition rate with 13.87%, signifying better employee retention. HR departments come in second with 19.5%, whereas sales are the highest at 30.56%. This suggests the company is having trouble retaining this department's staff.



Recommendations

- Leverage the best practices from the R&D department and utilize them for retention efforts.
- The sales department requires targeted interventions and focused strategies, improvements in the work environment, salary incentives to retain employees, and a proactive approach toward the problem.

3.4 Average Salary of Employees Who Left and Stayed



The average salary of those who stayed with the company stands to be $6,825. Whereas for those who left it was $5154, which was visibly lower than the former. This indicates that salary can be a factor affecting attrition. (Sorn et al., 2023) in their review study, came to the conclusion that “compensation plays a pivotal role in retention, and the employees with compensation dissatisfaction are more inclined towards quitting their jobs.” This narrative resonates with our findings.

Recommendations

- The difference between the two averages suggests that those with lower salaries are more likely to leave; hence, compensation can be used as a motivating factor for retaining employees.
- Market analysis is needed to ensure salaries meet industry standards.

- Create role-specific salary bands and inform employees to build trust and reduce apparent disparities.
- Payroll adjustments should be made for high-attrition roles and departments.
- Performance-based bonuses, retention bonuses, etc. should be implemented to reduce attrition.
- Professional development opportunities like paid training, certifications, and career advancement tie wage increases to promotions and skill growth.
- Analytics should be used for target-driven interventions and predictive analytics to identify departments and employees at risk of leaving.

4 CONCLUSION

Attrition, if not dealt with timely, is a long-term expense and can deteriorate the company's health, finances, success, and productivity. By the results obtained through our analysis, we have made the necessary recommendations for containing the attrition rate. A proactive approach would be the best mindset for dealing with attrition in the long run.

5 REFERENCE

- Dr. L.KESAVAN, M.S.D. (2022). A Study On Causes Of Employee Attrition. *Journal of Pharmaceutical Negative Results*, [online] pp.479–483. doi:https://doi.org/10.47750/pnr.2022.13.S08.62.
- Sankararaman Ganapathyraman, Ragothaman Cb, Ri, D.B. and Balakrishnan P (2023). A Study on Understanding Employee Attrition-Causes and Consequences in the Workplace. *ResearchGate*, [online] 13(4), pp.1004–1013. Available at: https://www.researchgate.net/publication/374813270_A_Study_on_Understanding_Employee_Attrition-Causes_and_Consequences_in_the_Workplace.
- Rajasekar, D. and Venkateswara Prasad, B. (2017). Article ID: IJMET_08_12_092-An Empirical Analysis. *International Journal of Mechanical Engineering and Technology (IJMET)*, [online] 8(12), pp.856–861. Available at: https://iaeme.com/MasterAdmin/Journal_uploads/IJMET/VOLUME_8_ISSUE_12/IJMET_08_12_092.pdf.
- Sorn, M.K., Fienena, A.R.L., Ali, Y., Rafay, M. and Fu, G. (2023). The effectiveness of compensation in maintaining employee retention. Open Access Library Journal, 10(7), pp.1–14. doi:https://doi.org/10.4236/oalib.1110394.